Asking RMBS loan data questions in plain English: results

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A language model turns each question into a small query spec, and a compiler turns the spec into SQL over governed metric definitions, so the model never does the arithmetic. These results are from a working prototype on synthetic data: 50 Australian RMBS deals, 21,945 loans and 773,336 loan-months in 2 trustee formats.

What is proved

check	result
Cleaning: canonical table against simulated truth	0 mismatches over 773,336 loan-months
Cleaning: resubmitted duplicates removed	721 of 774,057 raw rows
Cleaning: clean-up calls coded as repayments, recovered	7 deals
Compiler: output against hand-written SQL	49 of 49 match, worst difference 1.6e-15
Compiler: queries that render to Oracle	49 of 49
Scale: slowest question at 27,840,096 loan-months	under 0.75 s with default threads

The traps a plain SQL writer falls into

trap	question	correct	naive mistake gives
Arrears units	Q27	0.66%	0.59%
Calls as prepayments	Q29	21.6%	100.0%
Calls as prepayments	Q30	18.3%	49.0%
Exit rows as live loans	Q32	9,975	10,572
Unweighted average	Q33	51.96%	50.92%
Averaging CPR (dropped, gap too small)	Q2 2025 CPR	18.064%	18.063%

Governed system against plain text-to-SQL

The 40-question eval compares the 2 on the same questions, but it hasn't been recorded in this copy, because a run needs an Anthropic API key and none is stored. It takes 2 commands, and this page then regenerates with the results:

```
$env:ANTHROPIC_API_KEY = "<your key>"
python scripts/run_eval.py --mode record
python scripts/build_writeup.py
```

